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SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Design and apply optimization techniques to solve real-world computing problems with given constraints. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Constraint Based Problem Solving

Code of Conduct:

I Rishiyesh K.S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.S. Rishiyesh

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COMPUTER NETWORKS

CO5 — ASSESSMENT TOOL 1

CONSTRAINT-BASED PROBLEM SOLVING

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QUESTION 1 — Rural Branch Connectivity

A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

GIVEN

- A multinational company is opening a new branch in a remote rural location.
- High-speed wired Internet is unavailable.
- The office must support cloud-based applications and video conferencing.
- Secure communication with headquarters is required.
- The connectivity budget is limited.
- Business operations should remain uninterrupted.
- The design must consider bandwidth limitations, reliability, cost and future scalability.

Constraint Analysis

The dominant constraints are the absence of high-speed wired infrastructure, the need for interactive cloud and video traffic, secure headquarters communication, limited budget and uninterrupted operation. Therefore, the solution must balance availability, application performance, security and cost rather than optimize only one factor.

Proposed Solution

A business-grade 4G/5G fixed-wireless connection should be used as the primary WAN wherever adequate coverage and capacity exist. It avoids the unavailable wired infrastructure and can provide suitable throughput for cloud applications and video conferencing. The service should be selected using measured signal quality, expected peak bandwidth, latency and data limits rather than advertised maximum speed alone.

For continuity, an independent backup connection such as satellite or another wireless provider should be added. A dual-WAN router or SD-WAN edge can monitor the primary path and automatically redirect essential traffic when it fails. This preserves critical connectivity without requiring the backup link to carry the entire workload during normal operation.

Networking Devices

The branch should use a wireless WAN CPE/modem, business router or SD-WAN edge, next-generation firewall, managed switch and wireless access points. This provides WAN access, routing and failover, security, scalable LAN connectivity and user access.

Device	Role	Constraint addressed
4G/5G CPE / modem	Terminates wireless WAN	Wired Internet unavailable
Business router / SD-WAN	Routing, failover and QoS	Reliability + bandwidth
Next-generation firewall	Filtering, VPN and threat protection	Secure HQ communication
Managed switch	LAN connectivity and VLAN support	Scalability + control
Wireless AP	Office wireless access	Flexible connectivity

Security and Headquarters Connectivity

A site-to-site IPsec VPN should connect the branch security gateway to headquarters. IPsec protects confidentiality and integrity across the public Internet. Firewall rules should follow least privilege, permitting only required business services, while management interfaces should be restricted to authorized administrators. The VPN should automatically re-establish after WAN failover.

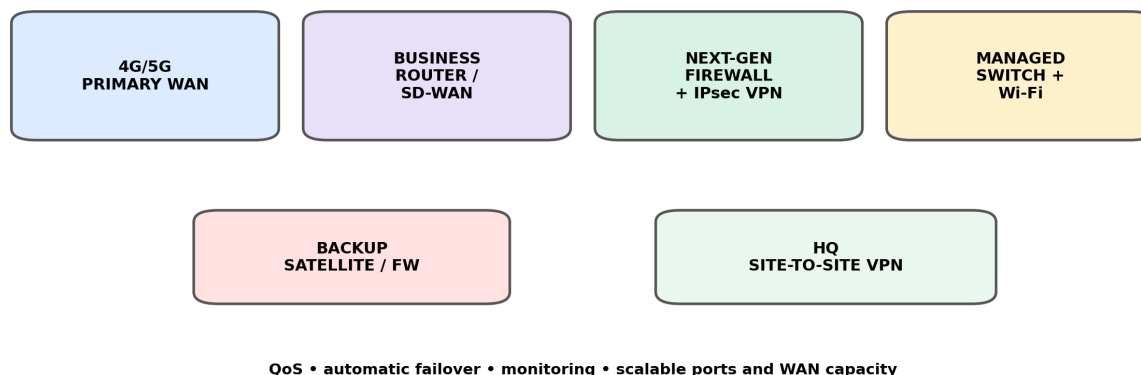
Bandwidth, Reliability, Cost and Scalability

QoS should prioritize video conferencing, voice and critical cloud applications over non-essential downloads and backups. Bulk transfers can be rate-limited or scheduled outside peak periods. WAN utilization, latency and packet loss should be monitored so capacity can be increased before service quality becomes unacceptable.

The architecture is scalable because WAN capacity, AP count and switch capacity can be increased independently. The main trade-off is variable wireless performance caused by signal and congestion; independent failover, QoS and monitoring reduce the operational effect of that limitation.

Recommended Architecture

RURAL BRANCH — RESILIENT CONNECTIVITY



Final Recommendation

The strongest balanced design is a primary business-grade 4G/5G wireless WAN with an independent backup link, dual-WAN/SD-WAN routing, QoS, a next-generation firewall and an IPsec VPN to headquarters. It directly satisfies the rural-access, reliability, bandwidth, security, budget and scalability constraints without unnecessary over-engineering.

QUESTION 2 — Campus-Wide Wireless Network for 3,000+ Students

A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

GIVEN

- The university must provide campus-wide wireless access for more than 3,000 students.
- Coverage is required in classrooms, laboratories, libraries and hostels.
- Many users may be connected simultaneously.
- Signal interference must be minimized.
- Access must be secure.
- Only a limited number of access points can be deployed because of budget constraints.
- The design must balance coverage, performance, security and cost.

Constraint Analysis

The key challenge is high-density capacity under a limited AP budget. A successful design therefore cannot rely on equal AP spacing alone. Placement, frequency selection and channel reuse must be based on user density and RF conditions so scarce AP capacity is used where it produces the greatest benefit.

Proposed Wireless Design

The campus should be divided into planning zones, with classrooms, laboratories and libraries receiving priority because they can contain many simultaneous active users. Hostels should be planned floor-by-floor while considering walls, floors and attenuation. A professional RF/site survey should determine final placement and identify interference.

5 GHz should be the preferred band for modern devices in dense areas because it generally offers more usable capacity and channels than 2.4 GHz. The 2.4 GHz band should remain available for longer reach and legacy devices but should be carefully channel-planned. If supported, 6 GHz can provide additional spectrum for compatible newer devices.

Band	Recommended role	Reason
5 GHz	Primary for dense users	Higher capacity and channel availability
2.4 GHz	Reach and legacy support	Longer propagation but greater interference
6 GHz, if supported	New compatible clients	Additional spectrum and reduced contention

Interference and Capacity Optimisation

Channel planning should avoid unnecessary overlap, and AP transmit power should be tuned so neighboring cells do not create excessive co-channel contention. Band steering can encourage capable clients toward higher-capacity bands. The objective is not maximum signal strength everywhere; it is stable coverage with sufficient capacity and controlled interference.

Because the number of APs is limited, deployment should prioritize locations where simultaneous demand is highest. If a weak area is discovered, placement, channel and power settings should be optimized before consuming another AP.

Security and Authentication

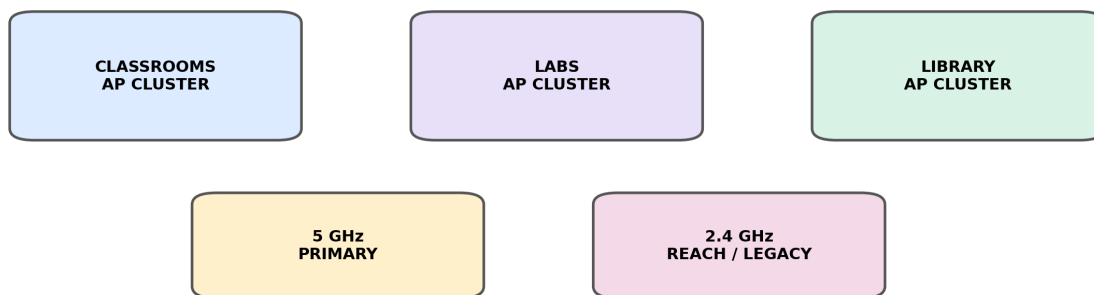
WPA3-Enterprise with 802.1X authentication backed by RADIUS is recommended. It provides individual identity-based authentication and scalable account management. Student, staff, guest and infrastructure traffic should be separated using VLANs and access-control policies so that access rights match the user's role.

Validation

The design should be validated using RSSI/SNR measurements, concurrent-client tests, throughput, latency, packet-loss and channel-utilization measurements. Authentication failures and roaming behavior should also be monitored. Results should be used to refine AP placement and RF settings.

Recommended Wireless Architecture

CAMPUS WI-FI — HIGH-DENSITY DESIGN



Survey-based AP placement • channel planning • WPA3-Enterprise + 802.1X/RADIUS

Final Recommendation

The optimized solution is a survey-driven, capacity-focused campus WLAN using 5 GHz as the preferred high-density band, controlled 2.4 GHz support, optional 6 GHz, careful RF planning and WPA3-Enterprise/802.1X authentication. It balances coverage, simultaneous-user performance, interference, security and the limited AP budget.

QUESTION 3 — Secure Financial Institution Network

A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

GIVEN

- The financial institution is upgrading its internal network.
- Cybersecurity must be improved.
- Employees need controlled access to sensitive customer information.
- Employees also need access to online banking services.
- Strict security policies and compliance requirements apply.
- Network performance should not be significantly reduced.
- Operational costs should not increase substantially.
- The design must include network segmentation, firewall deployment, authentication and intrusion detection.
- Security, performance, compliance and operational cost must all be considered.

Constraint Analysis

The institution requires strong protection without making the network impractical or excessively expensive. The most effective approach is defence in depth: isolate trust zones, control traffic between them, strengthen identity verification, detect suspicious activity and centrally manage security evidence.

Proposed Security Architecture

Use VLANs and Layer-3 security boundaries to separate employee workstations, internal application servers, sensitive customer-data systems, management systems and public-facing services. Public online-banking services should be placed in a DMZ. This prevents an Internet-exposed service from sharing an unrestricted security domain with sensitive databases.

Zone	Typical systems	Security objective
User VLANs	Employee workstations	Permit only required services
Server VLAN	Internal applications	Restrict application flows
Sensitive data zone	Customer/financial databases	Highest restriction and monitoring
DMZ	Public-facing services	Isolate Internet exposure
Management VLAN	Network/security administration	Authorized administrators only

Firewall Deployment and Access Control

A next-generation firewall should protect the Internet edge and enforce stateful policies between relevant security zones. Rules should follow least privilege: deny unnecessary traffic and explicitly permit approved services, destinations and protocols. VPN access should use strong authentication and controlled authorization.

Authentication and Authorization

Centralized identity management should be used for employees, with multi-factor authentication for privileged and sensitive operations. Role-based access control should map permissions to job responsibilities so employees can

access the customer information and online services required for their work, but not unrelated systems. Separate administrator accounts and restricted management interfaces further reduce privileged-access risk.

Intrusion Detection and Monitoring

IDS/IPS should monitor the Internet edge and important internal boundaries for suspicious or malicious activity. Security events should be centrally logged so repeated attacks, policy violations and possible lateral movement can be correlated. The firewall enforces access policy, while IDS/IPS adds visibility and detection.

Performance, Compliance and Cost Optimisation

Security controls should be applied according to risk rather than forcing identical deep inspection on every flow. Appropriate hardware capacity and monitoring of throughput, latency, CPU and memory help prevent security controls from becoming a bottleneck.

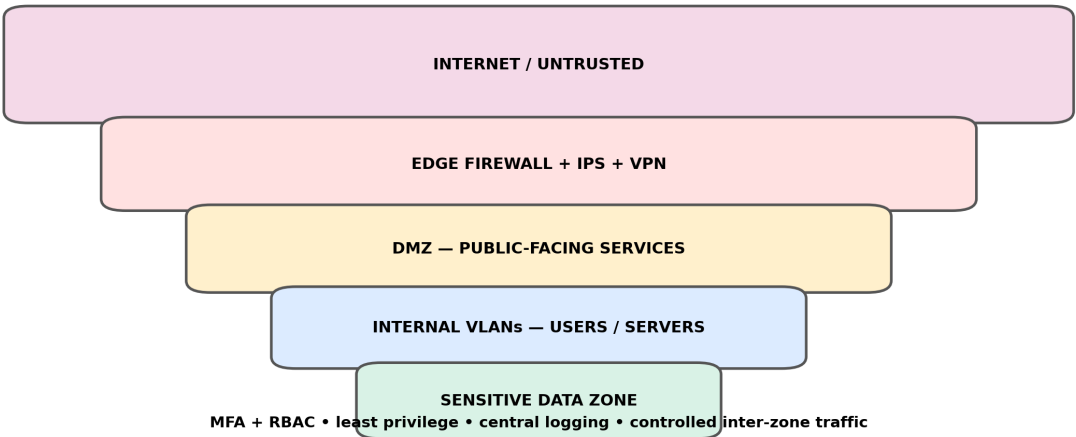
Centralized policy and log management reduces repetitive administration. Integrated security functions can be used where they meet the required risk level, avoiding unnecessary appliance duplication. Segmentation, least privilege, strong authentication, controlled administration and auditable logs provide a practical foundation for compliance.

Validation

Testing should confirm that unauthorized inter-VLAN traffic is blocked while approved banking and employee applications remain available. MFA and role permissions should be tested using representative accounts. Controlled security events should confirm IDS/IPS detection and logging, and external-reachability testing should verify that sensitive databases cannot be directly accessed from untrusted networks.

Recommended Defence-in-Depth Architecture

FINANCIAL INSTITUTION — DEFENCE IN DEPTH



Final Recommendation

The strongest feasible design combines segmentation, a controlled DMZ, next-generation firewalls, MFA, role-based authorization and IDS/IPS monitoring. It provides multiple independent security layers while preserving performance through selective inspection and capacity monitoring. Centralized management and logging reduce operational overhead and provide evidence that security and compliance controls are being enforced.